
Digital Signal Processing

Lecture 3

Difference Equation

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Difference Equation

- For all computationally realizable LTI systems, the input and output satisfy a difference equation of the form

$$\sum_{k=0}^N a_k y[n-k] = \sum_{k=0}^M b_k x[n-k]$$

- This leads to the recurrence formula

$$y[n] = -\sum_{k=1}^N \left(\frac{a_k}{a_0} \right) y[n-k] + \sum_{k=0}^M \left(\frac{b_k}{a_0} \right) x[n-k]$$

which can be used to compute the “present” output from the present and M past values of the input and N past values of the output

Linear Constant-Coefficient Difference(LCCD) Equations

Consider LTI systems satisfying

$$\sum_{k=0}^N a(k)y(n-k) = \sum_{k=0}^M b(k)x(n-k) \quad \text{ARMA}$$

Particular cases:

$$y(n) = \sum_{k=0}^M b(k)x(n-k) \quad \text{MA}$$

$$\sum_{k=0}^N a(k)y(n-k) = x(n) \quad \text{AR}$$

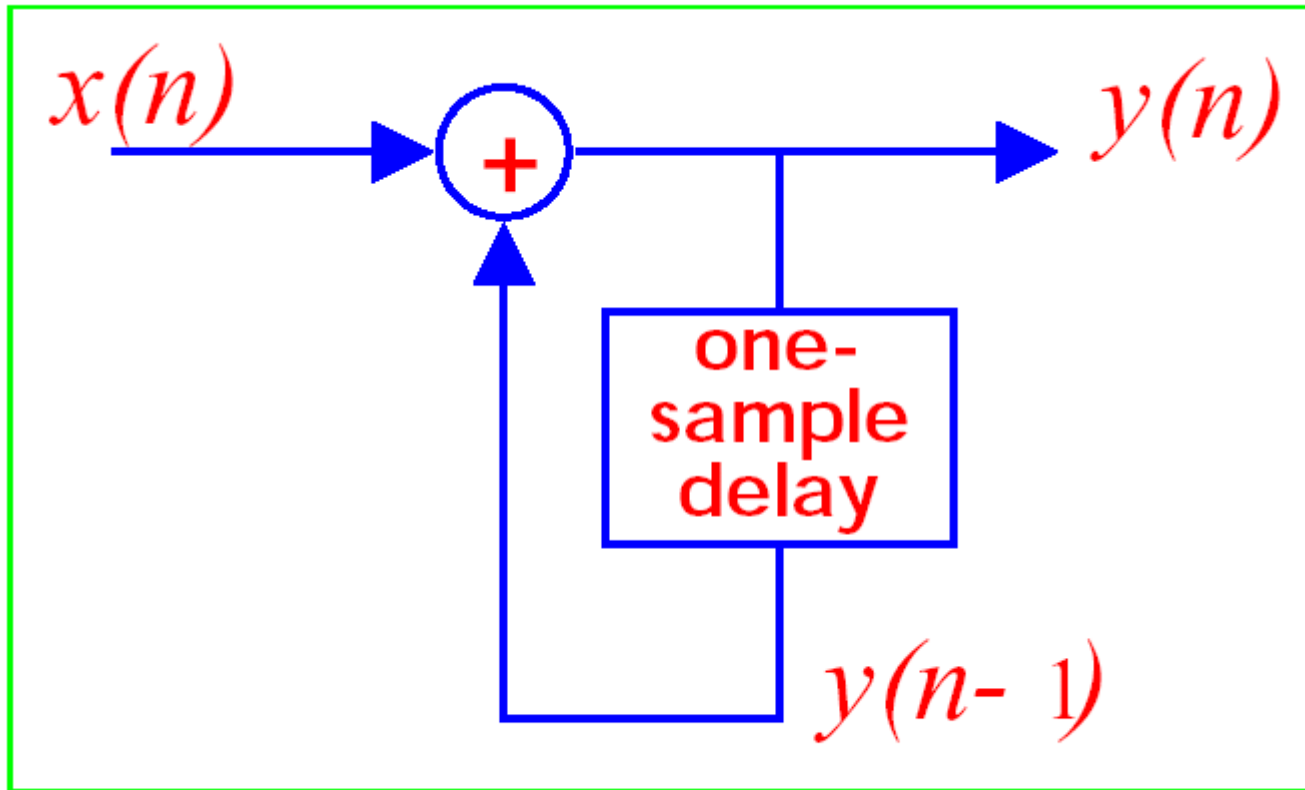
Linear Constant-Coefficient Difference (LCCD) Equations...(Continued)

Example:

$$y(n) = \sum_{k=-\infty}^n x(k) \quad \text{accumulator}$$

$$\begin{aligned} y(n) - y(n-1) &= \sum_{k=-\infty}^n x(k) - \sum_{k=-\infty}^{n-1} x(k) \\ &= x(n) + \left\{ \sum_{k=-\infty}^{n-1} x(k) - \sum_{k=-\infty}^{n-1} x(k) \right\} \\ &= x(n) \end{aligned}$$

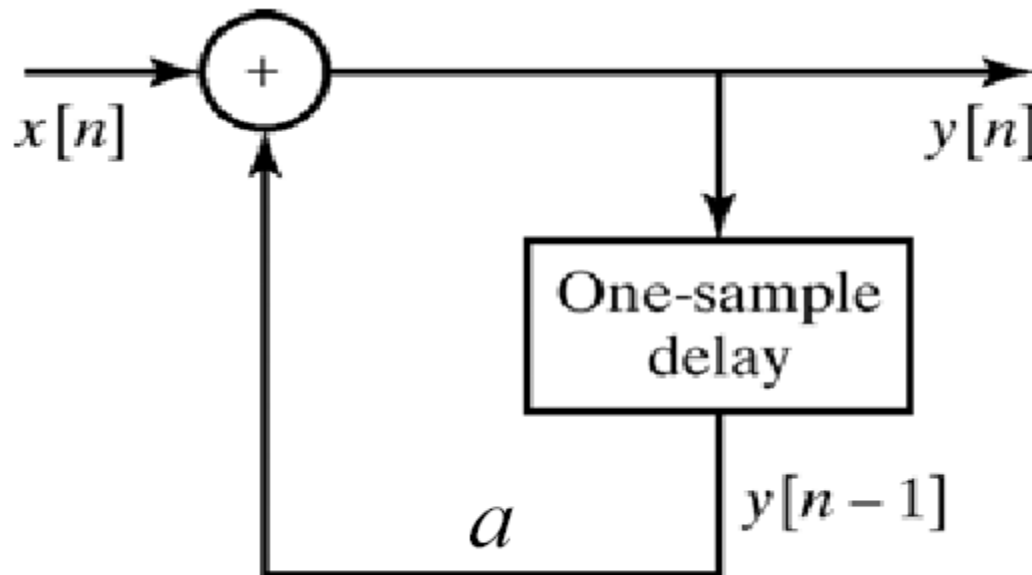
Linear Constant-Coefficient Difference (LCCD) Equations....(Continued)



First-Order Example

- Consider the difference equation
$$y[n] = ay[n-1] + x[n]$$

We can represent this system by the following block diagram:

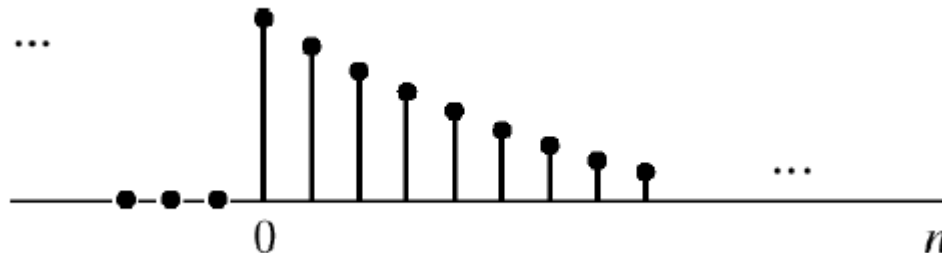


Exponential Impulse Response

- With initial rest conditions, the difference Equation has impulse response

$$y[n] = ay[n-1] + x[n]$$

$$h[n] = a^n u[n]$$



Linear Constant-Coefficient Difference (LCCD) Equations....(Continued)

Property: MA systems are bounded-input bounded-output (BIBO) stable, i.e.

$$|y(n)| = \left| \sum_{k=0}^M b(k)x(n-k) \right| \leq \sum_{k=0}^M |b(k)| |x(n-k)| < \infty$$

for any bounded input $|x(n)| < \infty$ and coefficient sequence $|b(n)| < \infty$.

Remark: AR systems may be unstable. For example, the system

$$y(n) = ay(n-1) + x(n)$$

is unstable for $a > 1$, because $y(n)$ is unbounded for bounded $x(n)$.

MATLAB and LTI Systems

»help conv

CONV Convolution and polynomial multiplication.

$Y = \text{CONV}(X, H)$ convolves vectors X and H . The resulting vector is length $\text{LENGTH}(X) + \text{LENGTH}(H) - 1$.

If X and H are vectors of polynomial coefficients, convolving them is equivalent to multiplying the two polynomials.

»help filter

FILTER One-dimensional digital filter.

$Y = \text{FILTER}(B, A, X)$ filters the data in vector X with the filter described by vectors A and B to create the filtered data Y . The filter is a "Direct Form II Transposed" implementation of the standard difference equation:

Digital Filter

$$Y = \text{FILTER}(B,A,X)$$

filters the data in vector X with the filter described by vectors A and B to create the filtered data Y . The filter is a "Direct Form II Transposed" implementation of the standard difference equation:

$$a(1)*y(n) = b(1)*x(n) + b(2)*x(n-1) + \dots + b(n_b+1)*x(n-n_b) - a(2)*y(n-1) - \dots - a(n_a+1)*y(n-n_a)$$

$$[Y,Zf] = \text{FILTER}(B,A,X,Zi)$$

gives access to initial and final conditions, Zi and Zf , of the delays.

LTI summary

$$y[n] = \sum_{k=-\infty}^{+\infty} x[k]h[n-k]$$

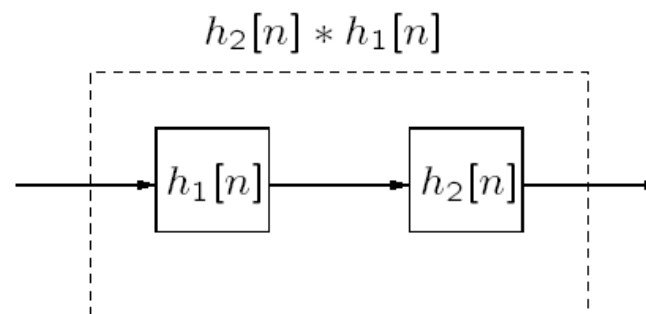
- Causality: $h[n] = 0 \ \forall n < 0$.

- Stability $\Leftrightarrow \sum_{n=-\infty}^{+\infty} |h[n]| < \infty$

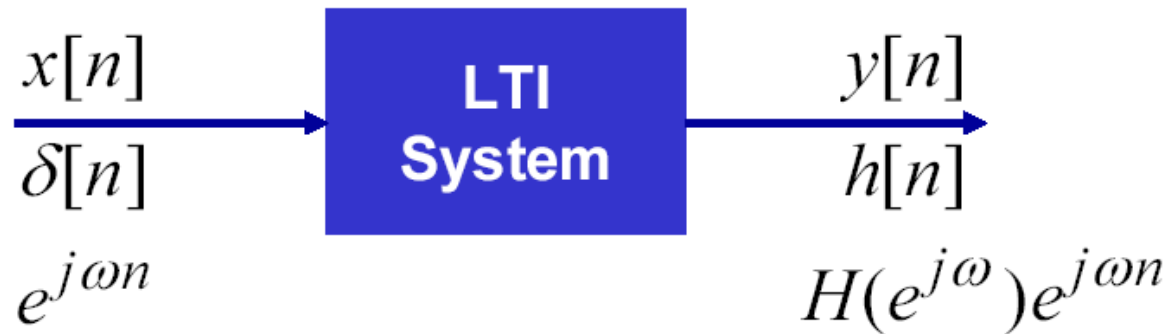
- FIR/IIR: length of $h[n]$ finite or not.

Note that an FIR system is always stable

- cascade connection:



Complex Exp Input Signal



$$y[n] = \sum_{k=-\infty}^{\infty} x[k]h[n-k] = \sum_{k=-\infty}^{\infty} h[k]x[n-k]$$

$$y[n] = \sum_{k=-\infty}^{\infty} h[k]e^{j\omega(n-k)} = \left(\sum_{k=-\infty}^{\infty} h[k]e^{-j\omega k} \right) e^{j\omega n}$$

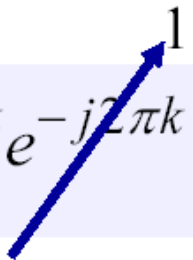
$$= H(e^{j\omega})e^{j\omega n}$$

Frequency response
 $H(e^{j\omega})$

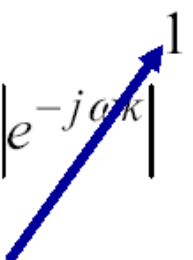
The Frequency Response

$$H(e^{j\omega}) = \sum_{k=-\infty}^{\infty} h[k]e^{-j\omega k}$$

- Periodicity of the frequency response:

$$H(e^{j(\omega+2\pi)}) = \sum_{k=-\infty}^{\infty} h[k]e^{-j(\omega+2\pi)k} = \sum_{k=-\infty}^{\infty} h[k]e^{-j\omega k} e^{-j2\pi k}$$


- Convergence of the frequency response:

$$|H(e^{j\omega})| = \left| \sum_{k=-\infty}^{\infty} h[k]e^{-j\omega k} \right| \leq \sum_{k=-\infty}^{\infty} |h[k]| |e^{-j\omega k}|$$


$$|H(e^{j\omega})| \leq \sum_{k=-\infty}^{\infty} |h[k]| < \infty$$

*Same as condition
for stability!*

Delay and First Difference

- Delay: $y[n] = x[n - n_d]$ $h[n] = \delta[n - n_d]$

$$x[n] = e^{j\omega n} \mapsto y[n] = e^{j\omega(n-n_d)} = \underbrace{e^{-j\omega n_d}}_{H(e^{j\omega})} e^{j\omega n}$$
$$H(e^{j\omega}) = e^{-j\omega n_d}$$

- First difference:

$$y[n] = x[n] - x[n - 1] \quad h[n] = \delta[n] - \delta[n - 1]$$

$$x[n] = e^{j\omega n} \mapsto y[n] = e^{j\omega n} - e^{j\omega(n-1)} = \underbrace{(1 - e^{-j\omega})}_{H(e^{j\omega})} e^{j\omega n}$$
$$H(e^{j\omega}) = 1 - e^{-j\omega}$$

More on the Ideal Delay

- The impulse response and frequency response of a pure delay are

$$h[n] = \delta[n - n_d] \Leftrightarrow H(e^{j\omega}) = e^{-j\omega n_d}$$

- Note that the magnitude of the frequency response is 1 for all ω

$$|H(\omega)| \equiv 1$$

and the phase is linear in ω

$$\arg\{H(\omega)\} = -\omega n_d$$

- Conversely, a frequency response with these characteristics corresponds to a pure delay

Discrete Time Fourier Transform

- Definition:

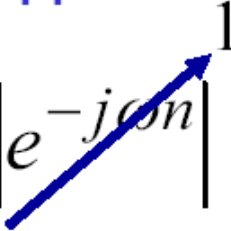
$$X(e^{j\omega}) = \sum_{n=-\infty}^{\infty} x[n]e^{-j\omega n}$$

Direct Transform

$$x[n] = \frac{1}{2\pi} \int_{-\pi}^{\pi} X(e^{j\omega})e^{j\omega n} d\omega$$

Inverse Transform

- Existence of the DTFT:

$$\left| X(e^{j\omega}) \right| \leq \sum_{n=-\infty}^{\infty} |x[n]| \left| e^{-j\omega n} \right| \leq \sum_{n=-\infty}^{\infty} |x[n]| < \infty$$


Properties

- Linearity
- time/frequency shifting :

$$\begin{aligned}x[n] &\xleftrightarrow{\mathcal{F}} X(e^{j\omega}) \\x[n-d] &\xleftrightarrow{\mathcal{F}} e^{-j\omega d} X(e^{j\omega}) \\e^{j\omega_0 n} x[n] &\xleftrightarrow{\mathcal{F}} X(e^{j(\omega-\omega_0)})\end{aligned}$$

- time reversal: $x[-n] \xleftrightarrow{\mathcal{F}} X(e^{-j\omega})$

Properties...(cont)

- differentiation: $nx[n] \xleftrightarrow{\mathcal{F}} j \frac{dX(e^{j\omega})}{d\omega}$
- Convolution: $x[n] * y[n] \xleftrightarrow{\mathcal{F}} X(e^{j\omega})Y(e^{j\omega})$
- modulation: $x[n]y[n] \xleftrightarrow{\mathcal{F}} \frac{1}{2\pi} \int_{-\pi}^{\pi} X(e^{j\omega'})Y(e^{j(\omega-\omega')})d\omega'.$ (Periodic convolution)
- Parseval theorem: $\sum_{n=-\infty}^{+\infty} |x[n]|^2 = \frac{1}{2\pi} \int_{-\pi}^{\pi} |X(e^{j\omega})|^2 d\omega$

Symmetric Properties

DTFT Properties:

$$x[-n] \Leftrightarrow X(e^{-j\omega})$$
$$x^*[n] \Leftrightarrow X^*(e^{-j\omega})$$

$$x[n] = x[-n] \Rightarrow X(e^{j\omega}) = X(e^{-j\omega})$$

even

even

$$x[n] = -x[-n] \Rightarrow X(e^{j\omega}) = -X(e^{-j\omega})$$

odd

odd

$$x[n] = x^*[n] \Rightarrow X(e^{j\omega}) = X^*(e^{-j\omega})$$

real

Hermitian symmetric

Consequences of Hermitian Symmetry

If $X(e^{j\omega}) = X^*(e^{-j\omega})$

then $\text{Re}[X(e^{j\omega})]$ is even

$\text{Im}[X(e^{j\omega})]$ is odd

$|X(e^{j\omega})|$ is even

$\angle X(e^{j\omega})$ is odd

And If $x[n]$ is real and even, $X(e^{j\omega})$ will be real and even

and if $x[n]$ is real and odd, $X(e^{j\omega})$ will be imaginary and odd

DTFT– Sinusoids

- Recall that

$$x[n] = 1 \Leftrightarrow X(e^{j\omega}) = \sum_{r=-\infty}^{\infty} 2\pi\delta(\omega + 2\pi r)$$

- Also note that

$$x[n]e^{j\omega_0 n} \Leftrightarrow X(e^{j(\omega - \omega_0)})$$

- Therefore

$$e^{j\omega_0 n} \Leftrightarrow \sum_{r=-\infty}^{\infty} 2\pi\delta(\omega - \omega_0 + 2\pi r)$$

$$\cos\omega_0 n \Leftrightarrow \sum_{r=-\infty}^{\infty} \pi\delta(\omega + \omega_0 + 2\pi r) + \pi\delta(\omega - \omega_0 + 2\pi r)$$

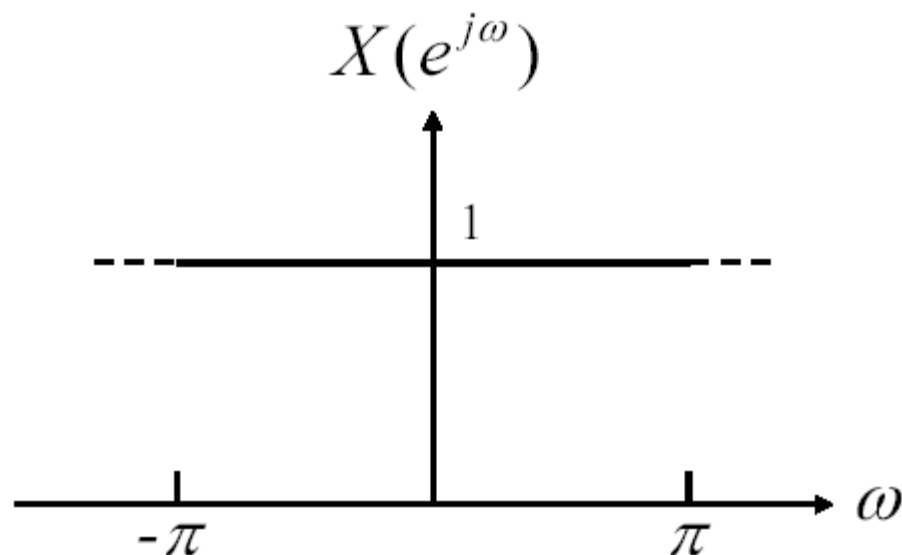
DTFT of Unit Impulse

Direct Transform

$$X(e^{j\omega}) = \sum_{n=-\infty}^{\infty} x[n]e^{-j\omega n} = \sum_{n=-\infty}^{\infty} \delta[n]e^{-j\omega n} = 1 \quad \text{for all } \omega$$

Inverse Transform

$$\delta[n] = \frac{1}{2\pi} \int_{-\pi}^{\pi} 1 \times e^{j\omega n} d\omega$$



Example

- Consider the real exponential signal

$$x[n] = a^n u[n]$$

- Its DTFT is obtained as follows:

$$X(e^{j\omega}) = \sum_{n=0}^{\infty} a^n e^{-j\omega n} = \sum_{n=0}^{\infty} (ae^{-j\omega})^n = \frac{1}{1 - ae^{-j\omega}}$$

$$X(e^{j\omega}) = \frac{1 - ae^{j\omega}}{(1 - ae^{-j\omega})(1 - ae^{j\omega})}$$

$$= \underbrace{\frac{1 - a \cos \omega}{1 + a^2 - 2a \cos \omega}}_{X_R(e^{j\omega})} + j \underbrace{\frac{-a \sin \omega}{1 + a^2 - 2a \cos \omega}}_{X_I(e^{j\omega})}$$

Magnitude and Angle Form

$$x[n] = a^n u[n] \quad \Leftrightarrow \quad X(e^{j\omega}) = \frac{1}{1 - ae^{-j\omega}}$$

$$X(e^{j\omega}) = |X(e^{j\omega})| e^{j\angle X(e^{j\omega})}$$

$$|X(e^{j\omega})|^2 = X(e^{j\omega}) X^*(e^{j\omega}) = \frac{1}{(1 - ae^{-j\omega})(1 - ae^{j\omega})}$$

$$= \frac{1}{1 + a^2 - 2a \cos \omega}$$

$$\angle X(e^{j\omega}) = \arctan\left(\frac{-a \sin \omega}{1 - a \cos \omega}\right)$$

Magnitude and Angle Plot

$$|X(e^{j\omega})| = \frac{1}{(1 + a^2 - 2a \cos \omega)^{1/2}}$$

$$\angle X(e^{j\omega}) = \arctan\left(\frac{-a \sin \omega}{1 - a \cos \omega}\right)$$

$$|X(e^{j\omega})| = |X(e^{-j\omega})|$$

$$\angle X(e^{j\omega}) = -\angle X(e^{-j\omega})$$

